

A template for a thesis written in Typst

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par

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acceptée sur proposition du jury:
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Lausanne, EPFL, 2026

To whatever...



Acknowledgements

Thanks to whomever.

Lausanne, 26 October 2026

J. D.

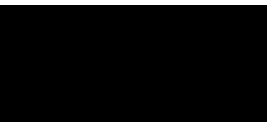


Abstract

This document is both a thesis template and its own manual.

Please place your very fancy summary here.

Key words: Typst, thesis template, notation index, matplotlib, EPFL.



Résumé

Ce document est à la fois un template et son mode d'emploi.

Ici, vous pouvez faire votre super résumé.

Mots-clés : Typst, template, index des notations, matplotlib, EPFL.

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1 Introduction

This template is the result of me writing an entire thesis in Typst. Features include:

- the **symbol index** of Chapter 2: declare a symbol once, use it a reference elsewhere, build an index symbol;
- **aligned equation grids** (Section 2.2), the equivalent of LaTeX's `subequations` inside an `align` environment;
- **figures generated by matplotlib as Typst markup** (Section 2.3), so that an axis label can reference an equation;
- a **CV** at the end of the thesis

1.1 A page of ordinary text

Chapters always open on a recto (odd, right-hand) page, so a blank verso before a chapter is intentional.

Lists:

- first item, on one line;
 - second item, enough text such that the new line wrap around, which respects the indent;
 - third item.
1. Numbered lists are styled by `itemize`, which indents the body of an item to the same place regardless of the width of its number.
 2. Second item.

Code is set in a monospaced face, inline like `EXAMPLE-CODE` or as a block:

```
cf = Config(typ=True, t_figsize=(5.4, 2.3))
cf.save("out/example_plot.typ")
```

You can also specify a language in an inside block with ````python main(a)``` : main(a) .`

1.2 Acronyms and citations

Acronyms are handled by `glossy` . Write `@MCF` and the first use expands to magnetic confinement fusion (MCF), while every later one is short: MCF again. The full list, with the pages where each acronym appears, is printed at the end of the document.

Other acronyms: International Thermonuclear Experimental Reactor (ITER), Joint European Torus (JET), magnetohydrodynamics (MHD).

Citations: `@helander_collisional_2005` gives (Helander and Sigmar 2005). You can cite a specific passage with `#cite(<helander_collisional_2005>, supplement: "Eq. (3.45)") : (Helander and Sigmar 2005, Eq. (3.45))`

2 Notation, equations and figures

2.1 The notation index

A symbol is **declared once**, at the place in the text where it is defined:

the distribution function `#symbol("Distribution function", $f_s$, key: "f_s")`

and used **everywhere else** through its key, with `r("f_s")`. Each use renders the registered symbol, links back to the declaration, and records the page. At the end of the document, Index of symbols lists every symbol with its definition page in bold and its uses in normal font.

So: let $f_s(x, v, t)$ be the distribution function of a species s , made of particles of mass m_s and charge e_s . It obeys the kinetic equation

$$\frac{\partial f_s}{\partial t} + \mathbf{v} \cdot \nabla f_s + \frac{e_s}{m_s} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \frac{\partial f_s}{\partial \mathbf{v}} = C_s[f_s], \quad (2.1)$$

where $\mathbf{B}$ is the magnetic field and C_s the collision operator.

Here is a reference Eq. (2.1). Write `#nref(<eq:kinetic>)` instead of `@eq:kinetic` when you only want the number alone (2.1).

2.1.1 Declaring versus using

- `symbol(description, math, key: "k")` is the **declaration**. It prints nothing on its own. It registers the symbol, drops an anchor for the links to land on, and adds a bold page entry. Put it immediately before the sentence that introduces the symbol. You can also declare a symbol which does not appear in the index, with `register: false` in the arguments.

- `r("k")` is a **reference** to the symbol. It prints the symbol, links to the declaration, and adds a normal page entry. You can change the appearance of a symbol with `display`, example: `#r("n_s", display: $n_s^2$) :  $n_s^2$ .`
- `nr("k")` links without adding an index entry.
- `n1r("k")` just prints the symbol, without link.

The `key` argument is optional. Without it, the key is guessed from the math content, but it is preferable to specify the key, as multiple symbols could conflict in a long thesis.

Symbols that appear with varying displays are easier to keep consistent as a function in `shared.typ`:

```
#let ns(..args) = r("n_s", display: $n_fm\sub(..args)$)
```

which gives n_s , n_e and $n_{i,0}$ from one declaration, all three referencing to the same place.

2.2 Aligned equations that share a number

The moments of f_s are:

$$n_s \equiv \int f_s d^3v \quad \text{Density} \quad (2.2a)$$

$$n_s \mathbf{u}_s \equiv \int \mathbf{v} f_s d^3v \quad \text{Fluid velocity} \quad (2.2b)$$

$$\frac{3}{2} n_s T_s \equiv \int \frac{m_s}{2} |\mathbf{v} - \mathbf{u}_s|^2 f_s d^3v \quad \text{Temperature} \quad (2.2c)$$

where T_s is the temperature in energy units. The block is Eqs. (2.2); a single line is Eq. (2.2c). Both are links.

The description column can be optional, e.g. `""`, but the five arguments per line are not.

2.3 A figure that references symbols

At thermal equilibrium the solution of Eq. (2.1) with the collision operator alone is the Maxwellian

$$f_M(v) = n_s \left(\frac{m_s}{2\pi T_s} \right)^{3/2} \exp \left(-\frac{v^2}{v_{th}^2} \right), \quad (2.3)$$

written in terms of the speed v and of the thermal speed $v_{\text{th}} = \sqrt{2T_s/m_s}$. A deuterium plasma at $T_D = 10 \text{ keV}$ has $v_{\text{th,D}} \approx 9.8 \cdot 10^5 \text{ m s}^{-1}$.

Figure 2.1 was made by `plots/plot_example.py`. It is **not** an image: matplotlib exported it as Typst markup, which the figure includes.

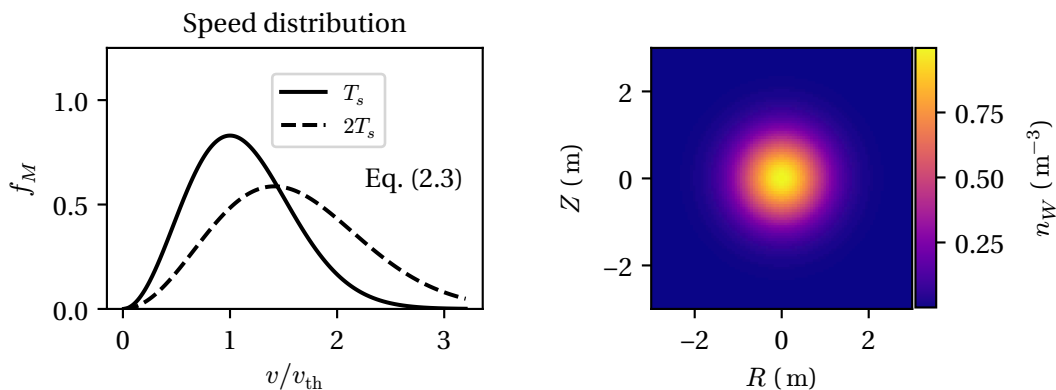


Figure 2.1 – Left: the speed distribution f_M of Eq. (2.3), at two temperatures. Right: some heatmap figure that doesn't slow down pdf readers.

The axis labels are linked to the document. Here I am defining symbols, that are referenced in the figure.

A note on floats

One can place a figure such as Figure 2.1 as a float with `placement: auto`, which lets Typst move it to a better place (w.r.t. the text). Typst 0.13 and 0.15 report a float's **source** position rather than the place it is displayed, so a reference to it would point at the wrong page. The template works around this by anchoring on the caption, which is laid out where the figure really is. That is why the template overrides the `ref` and `outline.entry` rules, and why a figure must have a caption for the workaround to work.

If you want `placement: auto` to stay within a sub-section (and not place figures outside their intended section), you can use the following trick:

```
// Flush floats before every level-2 heading (subsection boundary)
#show heading.where(level: 2): it => {
  place.flush()
  it
}
```


3 Conclusion

Happy writing.

A Incredible appendix

Reference: Appendix A. Equation below: Eq. (A.1).

Equation:

$$\int_0^\infty dx x^n e^{-x^2} = \frac{1}{2} \Gamma\left(\frac{n+1}{2}\right), \quad n > -1, \quad (\text{A.1})$$

which for the moments of Eqs. (2.2) gives the normalisation of f_M ,

$$\int d^3v f_M = n_s.$$

That one is written with `nonum`.

Index of symbols

Every symbol used in more than one place, with the page that **defines** it in bold and the pages that use it in normal weight. Runs of consecutive pages are merged into a range. All of it is generated – the list below is exactly the set of `symbol` declarations and `r` uses in the document.

Ordered by name

C

Collision operator C_s **3**

D

Density n_s **4, 5, 9**

Distribution function f_s **3, 4**

M

Magnetic field B **3**

Major radius coordinate R **5**

Maxwellian distribution f_M **4, 5, 9**

P

Particle charge e_s **3**

Particle mass m_s **3, 4, 5**

S

Species label s **3, 4**

Speed v **4, 5**

T

Temperature T_s **4, 5**

Thermal speed v_{th} **4, 5**

V

Vertical coordinate Z **5**

Ordered by symbol

B

B - Magnetic field 3

C

C_s - Collision operator 3

E

e_s - Particle charge 3

F

f_M - Maxwellian distribution 4, 5, 9

f_s - Distribution function 3, 4

M

m_s - Particle mass 3, 4, 5

N

n_s - Density 4, 5, 9

R

R - Major radius coordinate 5

S

s - Species label 3, 4

T

T_s - Temperature 4, 5

V

v - Speed 4, 5

v_{th} - Thermal speed 4, 5

Z

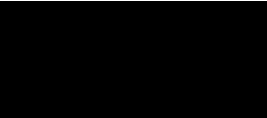
Z - Vertical coordinate 5

List of acronyms

HPC. high performance computing	
ITER. International Thermonuclear Experimental Reactor	2
JET. Joint European Torus	2
LCFS. last closed flux surface	
MCF. magnetic confinement fusion	2
MHD. magnetohydrodynamics. Single-fluid model of a plasma	2

Bibliography

Helander, Per, and D. J. Sigmar. 2005. *Collisional Transport in Magnetized Plasmas*. Digitally printed 1st pbk. version. Cambridge Monographs on Plasma Physics, no. 4. Cambridge University Press.



Postface

This here is not really necessary. Include this if you want to.

Funding

Bla bla bla.

Acknowledgement of computing resources

Bla bla.

Studies

- 2022 – 2026** **PhD in plasma physics, EPFL, Lausanne**
A template for a thesis written in Typst
- 2018 – 2021** **Master in physics, EPFL, Lausanne**
Master project on kinetic theory. Overall grade: 6 / 6

Conference contributions

- July 2026** **1000th EPS Conference on Plasma Physics, Poster, The Moon**
One line on what the poster showed

Work experience

- 2022 – 2026** **Doctoral assistant, SPC Theory, EPFL, Lausanne**
Research, drinking, and tours of the facilities for visitors

Notable projects

- 2022 – 2026** **EXAMPLE-CODE**
What the code does, in two lines, and where it lives.
github.com/example/example-code

General knowledge

- Plasma physics** Kinetic theory, post-neoclassic transport, magnetohydrodynamics voodoo.
- Parallel computing** OpenMP, PhD Graduate descent

Programming

- Advanced** C++, Python, Fortran, git
- Intermediate** Matlab, Klingon

Languages

- Native** Romansch
- Fluent** English (C1), LinkedIn Speak (B2)